

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	B.Tech AI/DS	Date:	

Code of Conduct

I Nirmal Kasi (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nirmal

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

1. Frame Semantics in Banking Customer-Service Systems

A digital banking platform uses semantic frames to understand customer requests. The NLP system identifies the **action, agent, object, and destination** involved in each request.

Customer Query	Generated Semantic Frame
"Transfer ₹20,000 from my savings account to my son's account."	TRANSFER(SourceAccount, Amount, DestinationAccount, Customer)
"Block my debit card immediately."	BLOCK(DebitCard, Customer)
"Send my transaction statement to my email."	SEND(TransactionStatement, Email, Customer)
"Increase my daily withdrawal limit."	MODIFY(WithdrawalLimit, Customer)

Additional transaction logs show:

Query ID	Actual User Intent	System Interpretation
B1	Transfer money to another account	Transfer money
B2	Block debit card	Block debit card
B3	Receive transaction statement by email	Send statement to email
B4	Increase withdrawal limit	Decrease withdrawal limit

Tasks:

1. Analyze the semantic frame of each banking request and identify the relationship between the action and its participants.
2. Identify the query in which the semantic interpretation is incorrect and explain why.
3. Analyze how incorrect identification of semantic roles can lead to an incorrect banking operation.
4. Recommend a frame-based semantic analysis approach to improve the reliability of banking conversational systems.

2. First-Order Predicate Calculus for Smart Agriculture

A smart agricultural system monitors crop fields using sensors and logical rules.

Field Data:

Field	Soil Condition	Irrigation Status	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

The system uses the following predicate rules:

- $Dry(x) \wedge IrrigationAvailable(x) \rightarrow NeedsWater(x)$
- $NeedsWater(x) \rightarrow Irrigate(x)$
- $Wet(x) \rightarrow \neg NeedsWater(x)$
- $IrrigationUnavailable(x) \rightarrow \neg Irrigate(x)$
- $Rice(x) \wedge NeedsWater(x) \rightarrow PriorityIrrigation(x)$

Tasks:

1. Represent the field observations using First-Order Predicate Calculus.
2. Using the given rules, determine which fields require irrigation.
3. Determine whether F1 and F3 should be treated differently by the automated irrigation controller.
4. Analyze whether predicate logic alone is sufficient for agricultural decision support when sensor information is incomplete or uncertain.

3. Word Sense Disambiguation in a Travel Recommendation System

A travel platform receives search queries containing ambiguous words. The recommendation engine uses surrounding words and user interactions to determine the intended meaning.

Search Query	Possible Sense 1	Possible Sense 2
"Amazon tour"	Amazon rainforest	Amazon company
"Safari booking"	Wildlife tour	Web browser
"Java vacation"	Java island	Java programming language
"Cruise package"	Ship journey	Cruise software/project

The system records the following user interactions:

Query	User Interaction
Amazon tour	Viewed rainforest trekking packages
Safari booking	Selected wildlife resort packages
Java vacation	Viewed hotels in Bali and Java
Cruise package	Viewed Mediterranean ship packages

However, another user searches: **"Safari download for Windows"** and clicks a browser download page.

Tasks:

1. Determine the intended sense of the ambiguous terms using the available context.
2. Identify the lexical and contextual cues that distinguish the possible senses.
3. Analyze why the same word may require different interpretations for different users.
4. Design an industrial-scale WSD strategy using query context, user history, embeddings, and click behaviour to improve travel recommendations.

4. Syntax-Driven Semantic Analysis in Legal Document Processing

A legal NLP system analyzes sentences from contracts and assigns semantic roles based on their syntactic structures.

The system produces the following analyses:

Legal Sentence	Parse Structure	Generated Semantic Interpretation
"The supplier delivered the equipment to the buyer."	Subject–Verb–Object	Supplier = Agent, Equipment = Object, Buyer = Recipient
"The buyer received the equipment from the supplier."	Subject–Verb–Object	Buyer = Agent, Equipment = Object, Supplier = Recipient
"The company terminated the contract after the violation."	Subject–Verb–Object	Company = Agent, Contract = Object
"The contract was terminated by the company."	Passive Construction	Contract = Agent, Company = Object

Tasks:

1. Analyze how syntactic structure, including active and passive constructions, influences semantic-role assignment.
2. Identify the incorrect semantic-role assignments in the given interpretations.
3. Explain how confusing grammatical subject with semantic agent can affect automated legal information extraction.
4. Recommend a syntax-driven semantic analysis architecture that can correctly handle active voice, passive voice, prepositional phrases, and long-distance dependencies in legal documents.

1. Semantic frames

A semantic frame represents the **action and the entities participating in that action**.

Customer Query	Semantic Frame	Roles
Transfer ₹20,000 from savings to son's account	TRANSFER(Source, Amount, Destination, Customer)	Customer = Agent, Savings = Source, ₹20,000 = Amount, Son's Account = Destination
Block my debit card	BLOCK(DebitCard, Customer)	Customer = Agent, Debit Card = Object
Send transaction statement to email	SEND(TransactionStatement, Email, Customer)	Customer = Agent, Statement = Object, Email = Destination
Increase daily withdrawal limit	MODIFY(WithdrawalLimit, Customer)	Customer = Agent, Withdrawal Limit = Object

2. Incorrect interpretation

B4 is incorrect.

Actual intent = Increase withdrawal limit

System interpretation = Decrease withdrawal limit

The system has reversed the **action direction**.

3. Effect of incorrect semantic roles

Incorrect role identification can cause the system to perform the wrong banking operation.

For example:

- Increase interpreted as decrease → withdrawal limit may be reduced.
- Wrong source → money may be taken from the wrong account.
- Wrong destination → money may be transferred to the wrong account.

4. Recommended approach

User Query



Intent Detection



Semantic Frame



Role Identification



Entity Extraction



Context Validation



Banking Operation

Important roles such as **action, source, destination, amount, customer, and object** should be validated before executing the transaction.

Conclusion: Frame-based semantic analysis reduces errors by explicitly connecting actions with their participants.

Q2. First-Order Predicate Calculus for Smart Agriculture

Given:

Field	Soil	Irrigation	Crop
F1	Dry	Available	Rice
F2	Wet	Available	Wheat
F3	Dry	Unavailable	Maize
F4	Wet	Unavailable	Rice

Rules:

$\text{Dry}(x) \wedge \text{IrrigationAvailable}(x) \rightarrow \text{NeedsWater}(x)$

$\text{NeedsWater}(x) \rightarrow \text{Irrigate}(x)$

$\text{Wet}(x) \rightarrow \neg \text{NeedsWater}(x)$

$\text{IrrigationUnavailable}(x) \rightarrow \neg \text{Irrigate}(x)$

$\text{Rice}(x) \wedge \text{NeedsWater}(x) \rightarrow \text{PriorityIrrigation}(x)$

1. Predicate representation

F1:

$\text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1}) \wedge \text{Rice}(\text{F1})$

F2:

$\text{Wet}(\text{F2}) \wedge \text{IrrigationAvailable}(\text{F2}) \wedge \text{Wheat}(\text{F2})$

F3:

$\text{Dry}(\text{F3}) \wedge \text{IrrigationUnavailable}(\text{F3}) \wedge \text{Maize}(\text{F3})$

F4:

$\text{Wet}(\text{F4}) \wedge \text{IrrigationUnavailable}(\text{F4}) \wedge \text{Rice}(\text{F4})$

2. Determine irrigation requirements

F1:

$\text{Dry}(\text{F1}) \wedge \text{IrrigationAvailable}(\text{F1})$

↓

$\text{NeedsWater}(\text{F1})$

↓

$\text{Irrigate}(\text{F1})$

↓

$\text{Rice}(\text{F1}) \wedge \text{NeedsWater}(\text{F1})$

↓

$\text{PriorityIrrigation}(\text{F1})$

So F1 requires irrigation with priority.

F2:

Wet(F2)
↓
¬NeedsWater(F2)
↓
No irrigation

F3:

Dry(F3)
but
IrrigationUnavailable(F3)
↓
¬Irrigate(F3)

F4:

Wet(F4)
↓
¬NeedsWater(F4)
↓
No irrigation
3. F1 vs F3

Yes, they must be treated differently.

F1 = Dry + Irrigation Available → Irrigate

F3 = Dry + Irrigation Unavailable → Do NOT irrigate

4. Is predicate logic alone sufficient?

No.

Predicate logic represents deterministic facts, but real sensor data may be:

- Incomplete
- Uncertain
- Noisy

Therefore, a practical agricultural system should combine predicate logic with **probabilistic or fuzzy reasoning**.

Conclusion: Predicate logic works well for clear sensor conditions, but uncertainty requires additional reasoning methods.

Q3. Word Sense Disambiguation in a Travel Recommendation System

The system receives ambiguous words such as **Amazon, Safari, Java, and Cruise**. User context and interactions help determine the intended meaning.

1. Intended senses

Query	Intended Sense	Evidence
Amazon tour	Amazon rainforest	Viewed rainforest trekking packages
Safari booking	Wildlife tour	Selected wildlife resort packages
Java vacation	Java island	Viewed hotels in Bali and Java
Cruise package	Ship journey	Viewed Mediterranean ship packages
Safari download for Windows	Safari browser	Clicked browser download page

2. Contextual clues

Amazon tour

tour + trekking
→ rainforest

Safari booking

booking + wildlife resort
→ wildlife safari

Java vacation

vacation + hotels
→ Java island

Cruise package

Mediterranean + travel package
→ ship journey

Safari download for Windows

download + Windows
→ browser/software

3. Why the same word has different meanings

The meaning of a word depends on:

- Surrounding words
- User history
- Interaction behaviour

For example:

Safari + booking
→ Wildlife Safari

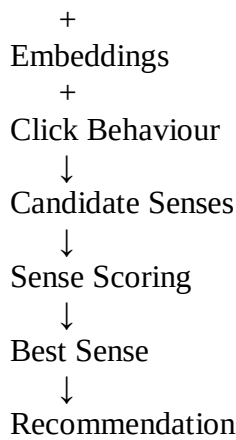
Safari + download + Windows
→ Safari Browser

4. Industrial-scale WSD strategy

Query Context

+

User History



Query context identifies surrounding clues.

User history identifies previous interests.

Embeddings measure semantic similarity.

Click behaviour provides evidence of actual user intent.

Conclusion: Combining linguistic context with behavioural and semantic information improves WSD accuracy and recommendation quality.

Q4. Syntax-Driven Semantic Analysis in Legal Document Processing

The system must determine semantic roles from syntactic structure. This is important because **the grammatical subject is not necessarily the semantic agent**, a distinction humans routinely manage without thinking and computers routinely screw up.

1. Active vs passive constructions

Active:

“The supplier delivered the equipment to the buyer.”

Supplier = Agent
Equipment = Object
Buyer = Recipient

Passive:

“The contract was terminated by the company.”

Company = Agent
Contract = Object

2. Incorrect semantic-role assignments

Sentence 2:

“The buyer received the equipment from the supplier.”

Given interpretation:

Buyer = Agent
Equipment = Object
Supplier = Recipient

This is incorrect.

Correct:

Buyer = Recipient
Equipment = Object
Supplier = Agent

Sentence 4:

“The contract was terminated by the company.”

Given interpretation:

Contract = Agent
Company = Object

This is incorrect because it is passive.

Correct:

Company = Agent
Contract = Object/Theme

3. Effect on legal information extraction

Confusing the grammatical subject with the semantic agent can cause the system to identify the **wrong person or organization as responsible for an action**.

In legal NLP, this can lead to incorrect extraction of:

- Obligations
- Responsibilities
- Actions
- Liabilities

4. Recommended architecture

Legal Document



Tokenization / POS Tagging



Syntactic Parsing



Active / Passive Detection



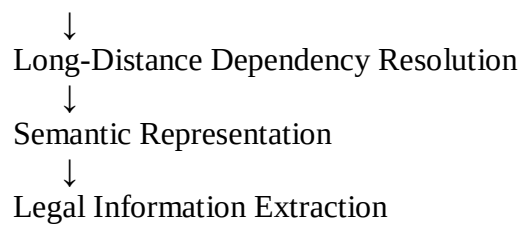
Dependency / Constituency Analysis



Semantic Role Labeling



PP Resolution



This architecture handles:

- Active voice
- Passive voice
- Prepositional phrases
- Long-distance dependencies

and therefore improves semantic interpretation of legal documents.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1	4	3	3	4	3	17	92
2	3	4	4	4	4	19	
3	4	4	4	4	3	19	
4	3	4	4	4	4	19	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____